higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

PROCESS CHEMISTRY
NQF LEVEL 2

(6050422)

13 November 2019 (Y-Paper)
13:00–16:00

This question paper consists of 6 pages

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a NEW page.
5. Use only blue or black ink.
6. Write neatly and legibly.

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QUESTION 1

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (1.1–1.5) in the ANSWER BOOK.

1.1 Personal protective clothes are used to safeguard the safety of workers against health and safety threats.

1.2 Scaling in industrial pipes results from water that has a lower pH.

1.3 Chemicals that are used to process living things are called agro-chemicals.

1.4 The South African constitution sets guidelines for matters that affect the environment.

1.5 Conditioning is the final step in the glass melting process.

(5 × 1) **[5]**

QUESTION 2

2.1 Petrochemicals are chemicals that are formed after prrocessing petroleum or crude oil.

2.1.1 Which South African company was formed in 1950 to produce petroleum from coal? (1)

2.1.2 Briefly explain why the company in QUESTION 2.1.1 above was formed. (3)

2.1.3 List THREE examples of substances that are derived from crude oil. (3)

2.1.4 Explain the difference between *coal* and *crude oil.* (2)

2.2 Coal is a non-renewable natural resource that is formed after millions of years.

2.2.1 What is a non-renewable natural resource? (1)

2.2.2 Name the South African utility company that uses coal to generate electrical energy. (1)

2.2.3 List FOUR elements that are the main components of coal. (4)

2.2.4 State FOUR industrial applications of coal. (4)

[19]

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QUESTION 3

3.1 Define the following terms:

3.1.1 Turbidity

3.1.2 Scaling

(2 × 2) (4)

3.2 Briefly discuss the polluters must pay principle. (3)

3.3 Mention FIVE steps in sequence of a typical waste water treatment. (5)

3.4 Draw a labelled diagram of an activated sludge digester. (7)

3.5 Briefly discuss flash mixing in water purification process. (3)

3.6 List SIX substances that are used as disinfectants. (6)

[28]

QUESTION 4

4.1 The main components of air are nitrogen and oxygen.

4.1.1 What is the % of oxygen in air? (1)

4.1.2 List FOUR physical properties of oxygen. (4)

4.1.3 What is the % of nitrogen in air? (1)

4.1.4 Name FOUR other gases (except oxygen and nitrogen) that are also found in air. (4)

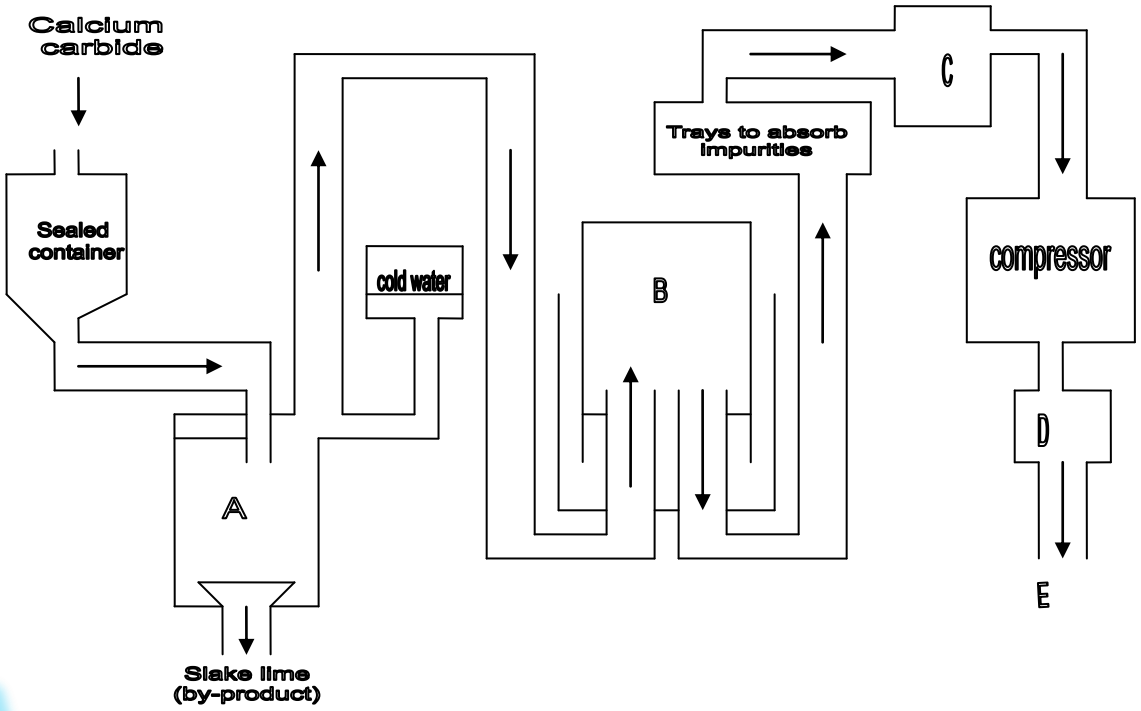
4.1.5 List FIVE industrial applications of nitrogen. (5)

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4.2 The following equation is used for the industrial preparation of acetylene from calcium carbide:

$$CaC_2\,(s) + H_2O\,(l) \rightarrow C_2H_2\,(g) + Ca(OH)_2\,(l).$$

Complete the following process flow diagram by filling in letters (A–E) in the ANSWER BOOK.



(5)
[20]

QUESTION 5

5.1 Choose a word or a term from the brackets to match a description given below. Write only the word or term next to the question number (5.1.1–5.1.5) in the ANSWER BOOK.

sodium sulphate; cullet; silica; feldspar; sodium carbonate

5.1.1 The main source of this compound is quartz sand and sandstones.

5.1.2 Used as a refining agent that helps to eliminate bubbles in glass.

5.1.3 Used as a fluxing agent during glass production process.

5.1.4 Recycled glass that is added to the raw materials.

5.1.5 Also known as soda ash.

(5 × 1) (5)

5.2 Name TWO properties of commercial glass. (2)

5.3 Draw a labelled flow diagram of glass production process. (12)
[19]

QUESTION 6

6.1 Indicate what the following personal protective equipment's are used for:

6.1.1 Respirator

6.1.2 Hardhat

6.1.3 Ear plugs

6.1.4 Face shield

6.1.5 Overalls

(5 × 1) (5)

6.2 Write a short description of an eye wash station. (4)
[9]

TOTAL: 100

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